

The exam is open book, open notes and open computer.

Upload your answers to gradescope by 12:50pm; or alternatively,  
submit the take home version, by noon on Monday 5 October.

All questions are worth the same amount of points.

1. Let  $Q = [(x_i, y_i), i = 0, 1, \dots, n-1]$  be a list of points, clockwise ordered, representing the vertices of a convex polygon, where two consecutive points in the list (modulo  $n$ ) represent an edge of the polygon.

Show that, for some  $k \in \{0, 1, \dots, n-1\}$ ,

$$Q = [(x_0, y_0), \dots, (x_{k-1}, y_{k-1}), (x_{k+1}, y_{k+1}), \dots, (x_{n-1}, y_{n-1})]$$

represents the list of vertices of a convex polygon with the  $k$ -th point removed.

2. Consider the status in the sweep line algorithm to intersect line segments.
  - (a) Explain a criterion to detect whether two adjacent segments in the status may intersect each other, without explicitly computing the intersection point.
  - (b) Make drawings to illustrate both cases: the case when the criterion excludes an intersection point, and the case when there may be an intersection point.
3. Let  $P$  be a list of clockwise ordered points, representing a nonconvex polygon. Suppose we want to compute the area of  $P$  as follows:
  - (a) compute the area of the convex hull  $Q$  of  $P$ ,
  - (b) triangulate the region defined by  $Q$  minus  $P$ ,
  - (c) subtract the area of  $Q$  from the sum of the areas of the computed triangles in  $Q$  minus  $P$ .

Do the following:

- (a) Define an algorithm to triangulate the region  $Q$  minus  $P$ .
  - (b) What is the cost of your algorithm?
4. Given any list of unordered points  $L$  and a point  $P$ . If  $P$  belongs to the convex hull of  $L$ , then  $P$  can be written as a convex combination of the points in  $L$ .  
Define a linear program to decide if  $P$  belongs to the convex hull of  $L$ .
  5. Assume we want to store data points  $(a, b)$  in a Kd tree where  $a$  is the age, and  $b$  the salary of an employee.  
Given that seniority and high salary are strongly correlated,  
how would you select the splitting lines to obtain a well balanced Kd tree?  
Illustrate your reasoning geometrically by drawings.

The questions below have to be read in the context of the questions with the same number on the previous two pages. In the take home version of the exam, good answers to the questions use software (CGAL, scipy, SageMath ...), or your own code.

All questions are worth the same amount of points.

1. Generate 20 random points, compute the convex hull, and then remove the middle vertex point. Relate the equations of the half planes that define the convex hull of the configuration with the removed point to the edges of the original convex hull.
2. Write code to implement a criterion to detect whether two line segments may or may not intersect. Illustrate the correctness of your code on the two cases.
3. Make a nonconvex polygon defined by random points on the unit circle, and alternately pushing the points inside and outside the unit disk. Thus, the convex hull will consist of the odd numbered points, as the even points are inside the unit disk.
  - (a) Use this test case to develop code for computing the area of the nonconvex polygon. Demonstrate the correctness of your code on regular polygons.
  - (b) Run sufficiently large examples to illustrate the cost.
4. Use a linear programming solver to decide if a point belongs to the convex hull of a given list of points or not.

Determine the cost of this check experimentally for a sequence of sufficiently large number of points.
5. Generate plausible data points, assuming a steady increase in salary with age.
  - (a) Compare your choice of splitting lines to construct a Kd tree, versus a random choice of splitting lines.
  - (b) Does your choice result in better balanced trees?